
CF07: COMPLETE FORMALISM — MEASUREMENT THEORY FOUNDATIONS FOR VDM: DECOHERENCE, EINSELECTION, AND BORN RULE EMERGENCE

Justin K. Lietz 
Neuroca, Inc.
justin@neuroca.ai

March 22, 2026

Abstract

Measurement is where the Void Dynamics Model (VDM) must reconcile a reversible “Hamiltonian limb” (J-only evolution) with the irreversible, entropy-producing “metric limb” (M-only evolution). This CF provides the measurement-theoretic bridge: (i) environment-induced decoherence as a concrete mechanism that suppresses interference in a dynamically selected pointer basis (einselection), (ii) finite-resolution coarse graining as an operational restriction that converts coherent superpositions into effectively classical mixtures, and (iii) a Born-rule mapping from quantum amplitudes to outcome frequencies derived from symmetry and compositionality constraints (without postulating collapse). The final deliverable is a falsifiable gate suite for CFN implementations: commutator residuals defining pointer stability, trace-distance decay quantifying decoherence, robustness under small coupling perturbations, and a Target M6 “Born frequency meter” requiring small Kullback–Leibler divergence between empirical frequencies and Born weights with explicit finite-sample bounds. The scope is deliberately restricted to measurable diagnostics that can be audited in code and tied to VDM gates.

Keywords measurement theory · decoherence · einselection · Born rule · coarse graining · quantum channels · Kullback–Leibler divergence · falsifiable gates · Void Dynamics Model

Contents

1	Objective and scope	3
2	Claim inventory and decisive falsifiers	3
3	Decoherence as a measurement channel	4
3.1	System–environment coupling and reduced dynamics	4
3.2	Decoherence factor and suppression of off-diagonals	4
3.3	Pointer states and einselection	5
3.4	Decoherence timescale as a threshold crossing	5
4	Born rule from symmetry and compositionality	5
4.1	Setup	5

4.2	Assumptions	6
4.3	Functional-equation derivation	6
5	Finite-resolution coarse graining and “classicality”	6
5.1	Windowed observables	6
5.2	Suppression of oscillatory interference terms	6
6	The $J \rightarrow M$ projection as bounded observation	7
6.1	Measurement as a CPTP map	7
6.2	Operational definition of the observation window	7
7	Target M6: Born frequency meter	7
7.1	Meter definition	7
7.2	Expected scaling and sample-size planning	8
7.3	High-probability bound (method of types)	8
7.4	Deterministic seeding and provenance	8
8	Robustness under perturbations	8
9	Gate suite summary	8
10	Integration map	9
11	Provenance and reproducibility	9
12	Limitations and future work	9

1 Objective and scope

This CF supplies the *measurement-theoretic* module needed to interpret the Void Dynamics Model (VDM) split between a reversible “Hamiltonian limb” and an irreversible “metric limb.” In practice, the split is operational: the J-only limb is the fine-grained, reversible dynamics, while the M-only limb is the coarse-grained dynamics inferred from bounded measurement. The central objective is to turn three foundational statements into *auditable observables* and *pass/fail gates*:

1. **Decoherence is real and measurable:** off-diagonal coherences in an operational pointer basis decay below a stated tolerance in a stated time window.
2. **The pointer basis is operationally defined (einselection):** a basis is “preferred” only if it is dynamically stable under the system–environment coupling, quantified by a commutator residual.
3. **Born weights map to frequencies as a meter:** outcome frequencies converge to Born probabilities with explicit finite-sample error bars, enabling a Target M6 “Born frequency meter” in the VDM unification program.

The scope is intentionally bounded:

- No philosophical collapse postulate is assumed; the framework treats measurement as a *channel* induced by coarse graining and environmental entanglement.
- No claim is made that decoherence alone *solves* all interpretational problems; the deliverable is a falsifiable mapping to diagnostics and gates.
- The only probability claim made is the one that can be directly tested: empirical frequencies versus Born weights under stated assumptions and tolerances.

The measurement module is classified as a *derived-limit* component tied to the VDM measurability axiom (A7) and its gate discipline [Lietz, 2025a,b,c].

2 Claim inventory and decisive falsifiers

This paper treats claims as *contracts* between a formal statement and a decisive metric. The four primary claims are:

Claim: C1 (Einselection / pointer stability) *Decisive metric:* $\max_i \|[H_{SE}, \Pi_i]\| \leq \varepsilon_{\text{ein}}$ for a stable pointer set $\{\Pi_i\}$.

A set of rank-1 projectors $\Pi_i = |i\rangle\langle i|$ is a pointer basis for the measurement channel only if the system–environment interaction Hamiltonian H_{SE} approximately preserves those projectors. Operationally: if no basis yields a small commutator residual, then “preferred basis” is *not* established for that mechanism [Zurek, 2003].

Claim: C2 (Decoherence / interference suppression) *Decisive metric:* trace distance to the dephased state falls below ε_{dec} within a stated time window.

Let $\rho_S(t)$ be the reduced system state and Δ_Π the dephasing map in the pointer basis. Decoherence is the statement that

$$D(t) \equiv \frac{1}{2} \|\rho_S(t) - \Delta_\Pi(\rho_S(t))\|_1 \leq \varepsilon_{\text{dec}}$$

for $t \geq \tau_{\text{dec}}$, where $\|\cdot\|_1$ is the trace norm. If $D(t)$ does not decay, the mechanism does not produce an operational measurement channel [Schlosshauer, 2004, 2007].

Claim: C3 (Born weights as frequencies) *Decisive metric:* $\text{KL}(f||p) \leq \varepsilon_{\text{KL}}$ at stated N , with stated confidence.

For a fixed state ρ and projective measurement $\{\Pi_i\}_{i=1}^d$, define Born weights

$$p_i \equiv \text{Tr}(\Pi_i \rho), \quad \sum_i p_i = 1.$$

Given N repeated trials, let $f_i = N_i/N$ be empirical frequencies. The meter asserts that f is statistically consistent with p in a quantitative sense (Target M6). Systematic deviations across repeated trials falsify the mapping for that mechanism [Masanes et al., 2019].

Claim: C4 (Information-theoretic bound) *Decisive metric:* $\mathbb{E}[\text{KL}(f||p)] \approx (d-1)/(2N)$ in the large- N regime.

Under i.i.d. sampling from p (explicitly assumed), the expected divergence between frequencies and the model scales as $\mathcal{O}(d/N)$, enabling a pre-registered sample-size plan for Target M6 [Kullback and Leibler, 1951, Wilks, 1938, Cover and Thomas, 2006].

The decisive falsifiers are the negations of the decisive metrics: failure to find a small commutator residual (C1), failure of trace-distance decay (C2), persistent KL deviations beyond finite-sample bounds (C3), or failure of the expected scaling collapse in d/N (C4).

3 Decoherence as a measurement channel

3.1 System–environment coupling and reduced dynamics

Consider a system S coupled to an environment E with total Hamiltonian

$$H_{\text{tot}} = H_S \otimes I_E + I_S \otimes H_E + H_{SE}. \quad (1)$$

For an initial product state $\rho_{SE}(0) = \rho_S(0) \otimes \rho_E(0)$, the unitary evolution is

$$\rho_{SE}(t) = U(t)\rho_{SE}(0)U(t)^\dagger, \quad U(t) = e^{-iH_{\text{tot}}t/\hbar}.$$

The operational state of the system is the reduced density operator

$$\rho_S(t) = \text{Tr}_E[\rho_{SE}(t)]. \quad (2)$$

This map $\rho_S(0) \mapsto \rho_S(t)$ is a completely positive trace-preserving (CPTP) channel; in Markovian limits it is generated by a Lindblad master equation [Lindblad, 1976, Schlosshauer, 2007].

3.2 Decoherence factor and suppression of off-diagonals

A standard decoherence mechanism appears when H_{SE} correlates distinguishable environment states with distinct system states [Zurek, 2003, Joos et al., 2003]. For a pure initial system state $|\psi\rangle = \sum_i c_i |i\rangle$ and a fiducial environment state $|E_0\rangle$, the joint state can take the form

$$|\Psi(t)\rangle = \sum_i c_i |i\rangle \otimes |E_i(t)\rangle, \quad (3)$$

leading to the reduced state

$$\rho_S(t) = \sum_{i,j} c_i c_j^* \langle E_j(t)|E_i(t)\rangle |i\rangle\langle j|. \quad (4)$$

Define the *decoherence factor*

$$\Gamma_{ij}(t) \equiv \langle E_j(t)|E_i(t)\rangle.$$

When $\Gamma_{ij}(t) \rightarrow 0$ for $i \neq j$, interference terms in the $|i\rangle$ basis are suppressed and $\rho_S(t)$ approaches the dephased (diagonal) mixture.

Lemma 1 (Diagonalization by orthogonal environment records). *Let $\rho_S(t)$ be as in Equation (4) and define the dephased state $\rho_{\text{diag}} = \sum_i |c_i|^2 |i\rangle\langle i|$. Then*

$$\frac{1}{2} \|\rho_S(t) - \rho_{\text{diag}}\|_1 \leq \sum_{i < j} |c_i| |c_j| |\Gamma_{ij}(t)|.$$

In particular, if $\max_{i \neq j} |\Gamma_{ij}(t)| \leq \gamma(t)$, then $D(t) \leq \gamma(t) \sum_{i < j} |c_i| |c_j| \leq \gamma(t) (d-1)/2$.

The lemma provides an auditable route: an implementation can measure $\Gamma_{ij}(t)$ (or directly $D(t)$) and compare to a tolerance.

3.3 Pointer states and einselection

A pointer basis is not a philosophical preference; it is an *operational stability condition*. Following the decoherence program [Zurek, 2003, Schlosshauer, 2007], the pointer projectors Π_i are those that (approximately) commute with the interaction term:

$$\|[H_{SE}, \Pi_i]\| \leq \varepsilon_{\text{ein}} \quad \text{for all } i. \quad (5)$$

This condition formalizes “the environment monitors an observable” and gives the decisive metric for C1.

Gate: G1 (Einselection residual) *Threshold:* $\max_i \|[H_{SE}, \Pi_i]\| \leq \varepsilon_{\text{ein}}$.

Implementation must report the operator norm (or a clearly stated proxy norm) of the commutator residual for the chosen pointer projectors. If no candidate basis yields residual below tolerance, the measurement-channel claim for that mechanism fails.

3.4 Decoherence timescale as a threshold crossing

Decoherence is typically exponential (or stretched-exponential) in simple dephasing models; the operational definition needed for CFN is the *threshold time*

$$\tau_{\text{dec}}(\varepsilon_{\text{dec}}) \equiv \inf\{t \geq 0 : D(t) \leq \varepsilon_{\text{dec}}\}. \quad (6)$$

A widely encountered scaling in thermal environments is that the dephasing rate increases with temperature and with (effective) coupling strength squared [Zurek, 2003, Schlosshauer, 2007]. For a simple order-of-magnitude estimate, write the interaction energy scale as

$$\Delta E \sim \lambda^2 k_B T,$$

where λ is a dimensionless effective coupling after extracting the thermal energy scale. The accumulated relative phase variance grows like $(\Delta E t / \hbar)^2$, giving the scaling estimate

$$\tau_{\text{dec}} \sim \frac{\hbar}{\lambda^2 k_B T} \quad (\text{scaling estimate; model-dependent prefactors omitted}). \quad (7)$$

The *paper does not require* the prefactor; the gate requires only that a stated mechanism yields a measured τ_{dec} consistent with its own stated model parameters.

Gate: G2 (Decoherence threshold) *Threshold:* $D(t) \leq \varepsilon_{\text{dec}}$ for all $t \geq \tau_{\text{dec}}$.

Implementation must report the trace-distance time series $D(t)$ and the threshold crossing time τ_{dec} for a declared ε_{dec} . Failure of decay below tolerance falsifies C2 for that mechanism.

4 Born rule from symmetry and compositionality

The goal is not to argue that “probability is fundamental.” The goal is to derive the unique mapping from amplitudes to outcome weights that is compatible with basic symmetry and composition requirements, and then to treat *frequency agreement* as the falsifiable observable. The operational derivation of the measurement postulates in Masanes et al. [2019] provides one rigorous route; the following derivation is included as a compact, auditable summary of the core constraints.

4.1 Setup

Consider a projective measurement with outcomes indexed by i and projectors Π_i . For a pure state $|\psi\rangle = \sum_i c_i |i\rangle$, define $x_i \equiv |c_i|^2$ so that $\sum_i x_i = 1$.

Assume there exists a function $F : [0, 1] \rightarrow [0, 1]$ such that the probability for outcome i depends only on x_i :

$$P(i|\psi) = F(x_i). \quad (8)$$

The assumptions below constrain F .

4.2 Assumptions

1. **Phase invariance:** $P(i|\psi)$ is invariant under $c_i \mapsto e^{i\theta} c_i$.
2. **Compositionality:** for independent systems with product state $|\psi\rangle \otimes |\phi\rangle$, joint probabilities factorize, implying $F(xy) = F(x)F(y)$.
3. **Normalization:** $F(1) = 1$ and $F(0) = 0$.
4. **Additivity for exclusive outcomes:** for a two-outcome measurement with amplitudes squared $(x, 1-x)$, one has $F(x) + F(1-x) = 1$.
5. **Regularity:** F is continuous (or measurable and bounded).

These assumptions are standard in operational reconstructions and are consistent with the structural postulates used by Masanes et al. [2019].

4.3 Functional-equation derivation

Theorem 1 (Born rule as the unique solution of the symmetry constraints). *Under the assumptions above, $F(x) = x$ for all $x \in [0, 1]$. Therefore $P(i|\psi) = |c_i|^2$.*

Proof sketch. Compositionality implies $F(xy) = F(x)F(y)$. With continuity and $F(1) = 1$, the solutions are $F(x) = x^\alpha$ for some $\alpha \geq 0$. Additivity $F(x) + F(1-x) = 1$ for all $x \in [0, 1]$ forces $\alpha = 1$. Hence $F(x) = x$. $\square$

The theorem yields an operational mandate: if a proposed measurement mechanism claims to implement projective measurements in basis $\{|i\rangle\}$, then observed frequencies should match $p_i = \text{Tr}(\Pi_i \rho)$ within finite-sample bounds. The derivation does not remove all interpretational questions, but it pins down the unique weight function compatible with basic symmetry and composition properties [Masanes et al., 2019].

5 Finite-resolution coarse graining and “classicality”

The decoherence channel suppresses interference in a pointer basis through entanglement with the environment. Independently, *finite measurement resolution* can wash out interference even without invoking detailed bath dynamics. The Kofler–Brukner framework makes this explicit: classical descriptions emerge when measurements are coarse compared to the interference scale [Kofler and Brukner, 2007].

5.1 Windowed observables

Let $A(x)$ be an observable with spatial dependence x . A detector with spatial resolution Δx effectively measures the coarse-grained observable

$$\tilde{A}(x) = \int_{\mathbb{R}} W_{\Delta x}(x - x') A(x') dx', \quad (9)$$

where $W_{\Delta x}$ is a normalized window function of width Δx (e.g. a top-hat or Gaussian). Expectation values then involve the convolution of oscillatory interference terms with $W_{\Delta x}$.

5.2 Suppression of oscillatory interference terms

Consider interference terms proportional to $e^{i\Delta k x}$ (coming from off-diagonal terms between wave numbers k_i and k_j). Under the window average,

$$\int_{\mathbb{R}} W_{\Delta x}(x) e^{i\Delta k x} dx = \widehat{W}_{\Delta x}(\Delta k).$$

For a top-hat window of width Δx , $\widehat{W}_{\Delta x}(\Delta k) = \text{sinc}(\Delta k \Delta x / 2)$. Therefore, when

$$\Delta k \Delta x \gg 1, \quad (10)$$

the windowed interference terms are strongly suppressed and only effectively diagonal contributions remain [Kofler and Brukner, 2007]. This provides a second, explicitly operational route to “classical mixtures” that can be turned into gates: given a declared Δx , an implementation can predict and measure the suppression factor.

6 The $J \rightarrow M$ projection as bounded observation

The VDM interpretation of measurement is that the metric limb (M-only) is the *effective* dynamics of bounded observation, not a fundamental replacement of reversible dynamics. In quantum language, “bounded observation” is represented by a coarse-graining CPTP map acting on the system state.

6.1 Measurement as a CPTP map

A standard measurement channel induced by unitary coupling to an environment (or apparatus register) and tracing out the unobserved degrees of freedom is

$$\mathcal{E}(\rho_S) = \text{Tr}_E[U(\rho_S \otimes \rho_E)U^\dagger]. \quad (11)$$

When $\mathcal{E}$ is a dephasing channel in the pointer basis, it can be written as

$$\mathcal{E}(\rho) = \sum_i \Pi_i \rho \Pi_i \quad (\text{ideal dephasing limit}).$$

Such channels are unital and typically increase the von Neumann entropy of the reduced state, reflecting information loss into the environment [Schlosshauer, 2007, Lindblad, 1976].

6.2 Operational definition of the observation window

For CFN implementation, an “observation window” is a kernel W summarizing finite resolution in space, time, and energy. One convenient representation is:

$$\rho_M = \int W(\xi) \rho_J(\xi) d\mu(\xi), \quad (12)$$

where ξ denotes the resolved degrees of freedom, $d\mu$ is the corresponding measure, and ρ_J denotes the fine-grained reversible state. The integral form emphasizes that the M-limb state is a *coarse average* of the J-limb description. In practice, Equation (11) and Equation (12) are two complementary representations of the same operational fact: part of the information is unobserved, and the effective state is obtained by marginalization.

7 Target M6: Born frequency meter

The VDM unification program defines target meters; Target M6 is the requirement that a measurement implementation reproduces Born frequencies with auditable finite-sample error control [Lietz, 2025c]. This section defines the meter as an algorithm plus gate suite.

7.1 Meter definition

Given:

- a state ρ (pure or mixed) on a d -dimensional Hilbert space,
- a projective measurement $\{\Pi_i\}_{i=1}^d$ (or more generally a POVM),
- N repeated trials producing outcomes $i_1, \dots, i_N$,

define Born weights and empirical frequencies by

$$p_i = \text{Tr}(\Pi_i \rho), \quad f_i = \frac{1}{N} \sum_{n=1}^N \mathbf{1}\{i_n = i\}.$$

Define the divergence metric in nats:

$$\text{KL}(f\|p) = \sum_{i=1}^d f_i \log\left(\frac{f_i}{p_i}\right), \quad (13)$$

with the usual convention $0 \log(0/p) = 0$.

Gate: G3 (Target M6: Born frequency) *Threshold:* $\text{KL}(f||p) \leq \varepsilon_{\text{KL}}$.

A Target M6 run passes if the observed frequencies satisfy $\text{KL}(f||p) \leq \varepsilon_{\text{KL}}$ at a declared N , across a declared seed suite, with provenance logged. A default engineering threshold used by VDM is $\varepsilon_{\text{KL}} = 10^{-3}$ (nats), but the meter is meaningful for any declared threshold [Lietz, 2025c].

7.2 Expected scaling and sample-size planning

In the large- N regime, $\text{KL}(f||p)$ admits a quadratic approximation around $f = p$. Using a second-order Taylor expansion and the multinomial covariance, one obtains the scaling estimate

$$\mathbb{E}[\text{KL}(f||p)] \approx \frac{d-1}{2N}, \quad (14)$$

consistent with Wilks-type asymptotics for likelihood ratios [Wilks, 1938, Kullback and Leibler, 1951]. This yields a conservative planning rule:

$$N \gtrsim \frac{d}{2\varepsilon_{\text{KL}}}.$$

For $\varepsilon_{\text{KL}} = 10^{-3}$, this suggests $N \gtrsim 500d$.

7.3 High-probability bound (method of types)

A complementary bound controls the probability that $\text{KL}(f||p)$ exceeds a threshold. For i.i.d. sampling from p on a d -symbol alphabet, a standard “method of types” bound gives

$$\Pr(\text{KL}(f||p) \geq \varepsilon) \leq (N+1)^d e^{-N\varepsilon}, \quad (15)$$

which is sufficient for practical confidence planning [Cover and Thomas, 2006]. In CFN practice, reporting both the realized KL and the bound in Equation (15) (with declared ε) provides an auditable statistical guardrail.

7.4 Deterministic seeding and provenance

Because Target M6 is a *meter*, reproducibility is part of the definition:

- A run must record the PRNG seed(s) and the exact sampling algorithm.
- Same seed $\Rightarrow$ same outcome sequence (bitwise determinism for the sampling stage).
- Different seeds $\Rightarrow$ different sequences but consistent statistics, verified by aggregation.

These requirements are consistent with the VDM gate discipline and tier standards [Lietz, 2025b,d].

8 Robustness under perturbations

A measurement mechanism that is so sensitive that infinitesimal changes in coupling destroy the pointer basis or the Born mapping is not a reliable instrument. Robustness is therefore a decisive falsifier.

Gate: G4 (Robustness / noncontextual stability) *Threshold:* *small perturbations to H_{SE} change p and Π by at most stated tolerances.*

Let $H_{SE} \mapsto H_{SE} + \delta H_{SE}$ with $\|\delta H_{SE}\|/\|H_{SE}\| \leq \eta$. The implementation must report the induced changes in (i) the chosen pointer projectors (e.g. via principal angles) and (ii) the Born weights p_i . Excessive instability falsifies the “instrument” interpretation for that mechanism.

9 Gate suite summary

For convenience, the CFN gate suite is summarized here:

Gate	Metric	Pass condition
G1	Einselection residual	$\max_i \ [H_{SE}, \Pi_i]\ \leq \varepsilon_{\text{ein}}$
G2	Decoherence threshold	$D(t) = \frac{1}{2} \ \rho_S(t) - \Delta_\Pi(\rho_S(t))\ _1 \leq \varepsilon_{\text{dec}}$ for $t \geq \tau_{\text{dec}}$
G3	Target M6 (Born frequency)	$\text{KL}(f\ p) \leq \varepsilon_{\text{KL}}$ with declared N and seed suite
G4	Robustness	perturbation stability under $\ \delta H_{SE}\ /\ H_{SE}\ \leq \eta$

Each gate must be accompanied by machine-readable artifacts (CSV/JSON) and provenance hashes consistent with the canon standards [Lietz, 2025b,d].

10 Integration map

This measurement module connects to prior CF components as follows:

- **Metriplectic split (CF01–CF02):** measurement motivates the operational need for a dissipative metric limb and connects it to coarse-grained channels.
- **Finite-speed causality (CF04):** bounded observation implies finite sampling and finite propagation constraints on what can be resolved as an “outcome.”
- **Integrability/closure discipline (CF05):** a measurement channel must not accidentally introduce hidden invariants; closure checks prevent silently constrained dissipation.
- **Information geometry (CF06):** Target M6 uses KL divergence and Fisher-type asymptotics; the statistical manifold viewpoint underlies the meter design.
- **Hierarchy program (CF03/A8):** hierarchical coarse graining is the natural multi-scale setting for repeated measurements and nested decoherence windows.

11 Provenance and reproducibility

Commit: 91332f4cf89a52d239eaab6659b8df8d8b9f3202 **Seed(s):** N/A **Artifacts:** Source: Derivation/Complete-Formalisms/CF07_Measurement_Theory_Decoherence_Born_Rule.md (SHA256: b01616073086f5490899f527c4e90ea2b8aaae21972bad8c20633d20bf12494d)

12 Limitations and future work

- **Model dependence of decoherence rates.** Equation (7) is a scaling estimate; concrete CFN notebooks must specify a bath model and verify the measured τ_{dec} against that model’s parameters.
- **Born rule assumptions must be explicit.** The symmetry/compositionality derivation is only as strong as its stated assumptions; alternative reconstructions should be compared, and any deviations must be caught by Gate G3.
- **Decoherence is not the entire story.** Decoherence explains suppression of interference in reduced states, but not the selection of a single outcome in a single trial; this CF does not claim otherwise [Schlosshauer, 2004].
- **Instrument design is the next step.** The gates here are the contract; implementing the contract in a validated CFN notebook (with artifact logging) is required before any “PROVEN” claims about specific physical regimes.

References

- Justin K. Lietz. VDM Canon: AXIOMS.md. GitHub repository snapshot, 2025a. A0–A8 axiom registry and gate discipline.
- Justin K. Lietz. VDM Canon: 00_VALIDATION_METRICS.md. GitHub repository snapshot, 2025b. Validation gates, tolerances, and logging formats.

- Justin K. Lietz. VDM Unification Program Spec: T0_Unification_Program_Spec_v1.md. GitHub repository snapshot, 2025c. Defines target meters (including Target M6) and the unification ladder.
- Wojciech H. Zurek. Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics*, 75(3):715–775, 2003. doi:10.1103/RevModPhys.75.715.
- Maximilian Schlosshauer. Decoherence, the measurement problem, and interpretations of quantum mechanics. *Reviews of Modern Physics*, 76(4):1267–1305, 2004. doi:10.1103/RevModPhys.76.1267.
- Maximilian Schlosshauer. *Decoherence and the Quantum-to-Classical Transition*. Springer, 2007. doi:10.1007/978-3-540-35775-9.
- Lluís Masanes, Thomas D. Galley, and Markus P. Müller. The measurement postulates of quantum mechanics are operationally redundant. *Nature Communications*, 10:1361, 2019. doi:10.1038/s41467-019-09348-x.
- Solomon Kullback and Richard A. Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951. doi:10.1214/aoms/1177729694.
- Samuel S. Wilks. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9(1):60–62, 1938. doi:10.1214/aoms/1177732360.
- Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2 edition, 2006. ISBN-13: 978-0471241959.
- Göran Lindblad. On the generators of quantum dynamical semigroups. *Communications in Mathematical Physics*, 48(2):119–130, 1976. doi:10.1007/BF01608499.
- Erich Joos, H. D. Zeh, Claus Kiefer, Domenico Giulini, Joachim Kupsch, and Ion-Olimpiu Stamatescu. *Decoherence and the Appearance of a Classical World in Quantum Theory*. Springer, 2 edition, 2003. doi:10.1007/978-3-662-05328-7.
- Johannes Kofler and Časlav Brukner. Classical world arising out of quantum physics under the restriction of coarse-grained measurements. *Physical Review Letters*, 99(18):180403, 2007. doi:10.1103/PhysRevLett.99.180403.
- Justin K. Lietz. VDM Canon: TIER_STANDARDS.md. GitHub repository snapshot, 2025d. Evidence ladder for PROVEN/PLAUSIBLE/NEEDS_DATA claims.